

**DESCRIPTION OF DESIGN DATA FOR
METALS: STRESS AND STRENGTH**

Section: **1.2 – METALLIC MATERIALS**
Chapter: **1250**
Title: **DESCRIPTION OF DESIGN DATA FOR
METALS: STRESS AND STRENGTH**
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1 INTRODUCTION

This chapter presents the definitions of the abbreviations and symbols used in following chapters to refer, in stress and strength analyses, to design data of metallic alloys in use.

1.1 DISCUSSION ON SOME GENERAL ASPECTS

The following discussion is temporally included in this chapter of the Stress Methods Manual. This inclusion herein is temporal; it shall be moved to a dedicated chapter in future revisions.

General

Materials Design Values to be used shall be consistent with applicable Stress Methods. In case a certain material property cannot be found in standard sources, or there are doubts about the applicability of some material data for a specific analysis, look for advice within your senior stress colleagues and/or the Stress Methods team.

Only materials fully qualified according to authorized specifications for aerospace applications will be allowed to be used in a part with any structural responsibility. For metallic materials and standard parts, this category includes the SAE/ASM specifications declared in the MMPDS, DIN/LN/WL specifications or Airbus specifications.

Statistical basis

General rules about the kind of statistical value of a material property that shall be used in analyses:

- Mean values of elastic moduli shall be used for internal loads/stress calculations
- B-basis values of ultimate allowable stresses shall be used for strength analyses. S-basis values shall be typically acceptable if B-basis values not available.

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However, applicable project regulations and certification/qualification basis may require taking into account special considerations. Use of A-basis material ultimate stresses, for example, shall be typically requested in strength analysis of single-load path structures of large aircraft under civil/commercial regulations (EASA CS-25), or in primary structure castings under military specifications (DEF-STAN 00-970).

Data sources

Valid sources of DV data: MMPDS, MMDH, HSB material data-sheets, Airbus qualification or Airbus Stress dedicated tests.

Inelastic behaviour

The inelastic behavior will be modelled using the well-known Ramberg-Osgood approximation, using the engineering values for stress and strain in calculations, except for non-linear FEM simulations where real/true values are typically needed by most authorized FEM solvers (contact with *Stress Methods & Opt.* dept. for confirmation).

Most sources of design values of mechanical properties of metallic materials, including those abovementioned referred as the authorized sources, provide these values in terms of engineering stress and strain.

The relationship between engineering and real/true stress and strain are the following (see MMPDS-9 chapter §1.8.2.h):

$$\epsilon_{true} = \ln(1 + \epsilon_{eng})$$

$$\sigma_{true} = \sigma_{eng} (1 + \epsilon_{eng})$$

Service conditions

Except for certain locations and extreme service conditions (e.g. close to the engine or any other heating sources), strength analyses shall be performed using material design values at room temperature. This policy shall be applicable for typical service conditions, from -55°C to 70°C.

Room temperature shall be the reference temperature for providing the material design values. In order to support structural analyses of cases at temperatures beyond abovementioned range (specially for high temperatures), and in addition to the coefficient of thermal expansion of the material, knock down factors at different temperatures shall be provided (as far as possible when experimental data were available). However, these factors, typically, will cover only short (up to ½ hour) material expositions to those ambient conditions, and do not account for the irreversible changes in material internal structure and performances caused by long term exposures to high temperatures. Notice that, for example, the temper of aluminum alloys can be affected by this phenomenon since 200°F / 90°C and above (see MMPDS figures 3.2.4.1.1(e)(f) or 3.7.5.2.1(e)(f)). Materials to be used in applications with long service in hot ambient shall need dedicated qualification programs to account for these effects.

Beyond the temperature topic, considerations regarding design values due to material exposition to other environmental factors (corrosion, embrittlement, etc...) shall not be treated here. Good design practices and selection of appropriate materials for the application should mitigate associated risks.

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Material direction

Mechanical properties of most metallic alloys are sensible to the direction of the sustained stresses. Material performance in L, LT directions are typically better than in S, ST directions. This phenomenon may be relevant for design of parts with complex multi-axial stresses, as fittings, and shall be accounted in analysis. Anyway, this anisotropy is normally not as significant in metal alloys employed as structural materials in aerospace application as it is in composite materials. The preferred approach in those cases with loaded metallic parts subjected to significant 3D effects shall be to conservatively select, as design value for strength, the minimum allowable value among the material directions.

Material designation / identification

Full designation of the material according to an authorized material specification shall be provided. This complete designation and the corresponding technical specification will permit to identify the material alloy (its chemical composition), its format (sheet, plate, forging, casting...), and the full thermal treatment used to get the material to its final state in the installed part.

Link with the source of the design data used in analyses shall have to be provided also.

2 DEFINITIONS OF DESIGN DATA FOR METALS (STRESS)

The following set of data shall be provided for each relevant material format/size, material direction and service temperature.

Physical properties

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
α	1/°C	Coefficient of thermal expansion
ρ	Kg/m ³	Density

Mechanical properties

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
E	MPa	Elastic Young's modulus
E _c	MPa	Compression modulus
G	MPa	Shear modulus, or Modulus of Rigidity
ν	-	Poisson's ratio
F _{cy}	MPa	Yield compression stress
F _{ty}	MPa	Yield tensile stress
F _{tu}	MPa	Ultimate tensile stress
n	-	Ramberg-Osgood exponent of the stress/strain curve in tension

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n_c	-	Ramberg-Osgood exponent of the stress/strain curve in compression
F_{su}	MPa	Ultimate shear strength
F_{bry}	MPa	Bearing yield strength in fastened joints, for ratios of edge distance to hole diameter equal to 1.5 and 2.0
F_{bru}	MPa	Ultimate bearing strength in fastened joints, for ratios of edge distance to hole diameter equal to 1.5 and 2.0
e	-	Elongation, i.e. the permanent plastic deformation after the tensile failure (typically in %)

Some other remarks and useful relationships for isotropic materials:

- Shear and Tension moduli: $G = E / [2 \cdot (1 + \nu)]$ (MMPDS chapter §1.4.6)
- Ramberg-Osgood exponent in tension: $n = \log(e / 0.2\%) / \log(F_{tu}/F_{ty})$
- Approximation of the Ramberg-Osgood exponent in compression if not explicitly available:

$$n_c = \min[n, \log(e / 0.2\%) / \log(F_{tu}/F_{cy})]$$
- Shear and Tension yield strength: $F_{sy} = 0.577 \cdot F_{ty}$ (*)
 (*) Relation derived from the classical Von Mises (octahedral) stress criterion for plasticity in tension and shear, i.e.: $\sigma_{vm}^2 = \sigma^2 + 3 \cdot \tau^2$
- Bearing strength of aluminum alloys in MMPDS tables will normally need a correction due to the “edgewise bearing” effect reported in MMPDS paragraph and table 3.1.2.1.1

3 DATASHEET TEMPLATE FOR A METALLIC MATERIAL

MATERIAL IDENTIFICATION

Name/designation	
Version	
Description	
Specifications	
Equivalences	
References	
Units	International System of Units (SI): subset of derived units (N,mm)

PHYSICAL PROPERTIES

Property	Symbol	Units	Value/s					
Density	ρ	Kg/m ³						
Reference temperature	T _{ref}	°C						
Coefficient of thermal expansion (CTE)	T	°C						
	α	1/°C						

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MECHANICAL PROPERTIES

Elastic moduli and Ramberg-Osgood exponents (at room temperature)

Property	Symbol	Units	Value
Young modulus	E	MPa	
Compression modulus	E_c	MPa	
Shear modulus	G	MPa	
Poisson's ratio	ν	-	

Material strengths (at room temperature)

Property	Symbol	Dir.	Units	Values								
Thickness (max.)			mm									
Basis			-	S	A	B	S	A	B	S	A	B
Ultimate tension stress	F_{tu}	L	MPa									
		LT	MPa									
		ST	MPa									
Yield tension stress	F_{ty}	L	MPa									
		LT	MPa									
		ST	MPa									
Yield compression stress	F_{cy}	L	MPa									
		LT	MPa									
		ST	MPa									
Ultimate shear stress	F_{su}	L-S	MPa									
		T-S	MPa									
		S-L	MPa									
Ultimate bearing stress	$F_{bru_1.5}$	L	MPa									
		LT	MPa									
		ST	MPa									
	$F_{bru_2.0}$	L	MPa									
		LT	MPa									
		ST	MPa									
Yield bearing stress	$F_{bry_1.5}$	L	MPa									
		LT	MPa									
		ST	MPa									
	$F_{bry_2.0}$	L	MPa									
		LT	MPa									
		ST	MPa									
Elongation	e	L	-									
		LT	-									
		ST	-									
RO exponent in tension	n	L	-									
		LT	-									
		ST	-									
RO exponent in compression	nc	L	-									
		LT	-									
		ST	-									

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Effect of temperature: Environmental Knock Down Factors

Property	Symbol	Units	Value/s					
Environmental Knock Down Factors for elastic moduli, Poisson's ratio, Strengths, elongation and Ramberg-Osgood exponents	T	°C						
	$EKDF_E$	-						
	$EKDF_Ec$	-						
	$EKDF_G$	-						
	$EKDF_v$	-						
	$EKDF_Ftu$	-						
	$EKDF_Fty$	-						
	$EKDF_Fcy$	-						
	$EKDF_Fsu$	-						
	$EKDF_Fbru$	-						
	$EKDF_e$	-						
	$EKDF_n$	-						
	$EKDF_nc$	-						

4 APPENDICES

OTHER ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
A-basis	-	Statistically-based material property set at the point where, with a 95% confidence, at least 99% of the population will exceed
B-basis	-	Statistically-based material property set at the point where, with a 95% confidence, at least 90% of the population will exceed
DV	-	Design Value, i.e. the value for a mechanical property of a structural material to be used in stress analyses
S-basis	-	Specification minimum, or value based on specification minimum, of a material property
TBC	-	To Be Confirmed
σ	MPa	Axial stress
τ	MPa	Shear stress

REFERENCES

	Document Reference	Issue	Title
Ref. 1	[SAE] MMPDS	9	Metallic Materials Properties Development and Standardization
Ref. 2	[IHS-ESDU] MMDH	-	Metallic Materials Data Handbook (formerly DEF-STAN 00-932)

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SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 05/03/2020	G. Baños	First issue	All pages
1.1 12/02/2021	G. Baños	Clarification regarding use of temperature knowck down factors and addition of template of DV's datasheet	Sections §1.1, §3

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Methods (TEPAP-TL4)</i>	G. Baños 12/02/2021	Florian Glock 12/02/2021

Table 3. *Approval signatures table for last revision of this chapter*